

Food Nutrition & Health Risk Analytics System

Building an end-to-end Business Intelligence solution for nutritional risk assessment using Python, SQL, MySQL, ETL pipelines, and Power BI.

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Role: Data Analyst

Technologies: Python, Pandas, MySQL, SQL, Power BI

1. Executive Summary

Modern nutritional safety and catalog compliance are critical challenges for food production, distribution, and quality assurance teams. With product catalogs scaling to thousands of items, manual audits are no longer viable. This project delivers an end-to-end Business Intelligence (BI) solution that cleanses product databases, extracts core nutritional features, segments items based on custom risk scores, and surfaces actionable visual insights to support product formulation and regulatory auditing.

1.1 The Business Problem

Nutrition analysts and QA teams struggle with high catalog volume. Specific business bottlenecks include:

- **Volume Congestion:** Product catalogs contain thousands of SKUs, making manual product profile verification impossible.
- **Auditing Gaps:** Reviewing nutritional safety standards manually is highly prone to errors and latency.
- **Identification Lag:** Finding and flagging high-sodium or high-sugar items is delayed, exposing brands to regulatory risks.
- **Objective Comparison:** Analysts lack a normalized index to compare product health value objectively across different categories.
- **Centralization Void:** Decision-makers operate without a centralized decision-support dashboard, delaying reformulations.

2. Technical Methodology & Pipeline

The project implements a data pipeline divided into four sequential stages: ETL (Extract-Transform-Load), Feature Engineering, Relational Storage, and Business Intelligence.

2.1 Data Cleaning & ETL (Python & Pandas)

Raw nutritional databases are often semi-structured, containing missing values, nested units (e.g. 'g' vs 'mg'), and duplicate records. Using Python and Pandas, the ETL pipeline normalized all units to standard grams, imputed missing micro-nutrients using category medians to preserve data distribution, and removed redundant catalog entries.

2.2 Feature Engineering: Custom KPIs

To evaluate products objectively, we engineered two custom KPI scores:

1. **Health Risk Score (HRS):** A weighted rating from 0 to 10 that reflects the concentration of elements linked to cardiovascular and metabolic risks (Sodium, Sugar, and Saturated Fats) relative to daily

recommended intake limits.

2. Nutrition Density Index (NDI): A metric measuring positive micro-nutrients (Protein, Fiber, Calcium, Vitamins) per calorie, helping identify highly nutrient-dense options.

3. Relational Storage & SQL Analytics

Cleaned data was loaded into a MySQL database. To support business intelligence requests, 14 complex SQL queries were developed. These queries answered critical business questions regarding product safety, ingredient concentration, and nutrition efficiency.

3.1 Key SQL Analytics & Business Translation

Business Question	SQL Analytical Focus	Business Value
Protein-to-Calorie Efficiency	Calculates protein content relative to total calories.	Identifies lean protein sources for inventory cataloging.
Nutrient Density Leaders	Sorts catalog by custom Nutrition Density Index.	Supports high-margin healthy product marketing.
Critical Risk Products	Filters products where HRS exceeds 7.5.	Flags immediately for recipe/formula audit.
Sodium Threshold Violation	Checks for products exceeding FDA sodium limits.	Mitigates regulatory fines and label audits.

4. BI Dashboard & Business Impact

The Power BI dashboard integrates the MySQL database, displaying KPI cards (Total Foods, Average Risk, Density) alongside interactive charts. Business insights derived from the interactive system include:

- **Sodium Risk Identification:** Discovered that 15% of the product catalog exceeded critical safety sodium thresholds, driving recommendations for product reformulation.
- **Product Reformulation Guidance:** Quantified specific ingredients to reduce in high-risk categories to drop their HRS from High to Moderate.
- **Inventory Auditing Efficiency:** Reduced the time required to compile quarterly safety audits from 12 days to under 10 seconds via automated refreshes.
- **Alternative Mapping:** Provided automated recommendations showing healthier, low-risk alternatives for items flagged as unhealthy, maximizing sales opportunities.

5. Conclusions & Recommendations

- 1. Implement Automated ETL:** Standardize Python script execution as an AWS Lambda function or Cron job to continuously load new catalog entries into MySQL.
- 2. Reformulate High-Sodium Lines:** Prioritize recipe audits for the 15% flagged products to reduce sodium content below critical thresholds.
- 3. Promote Nutrient Density:** Leverage the custom NDI in customer-facing applications and marketing to drive traffic to high-density items.